

Kousumi Mondal

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PROFESSIONAL SUMMARY

Final-year Computer Science Engineer specializing in Agentic AI, Retrieval-Augmented Generation, and production-grade ML systems engineering. Experienced in architecting full-stack AI platforms — spanning deep learning model design, scalable RAG pipelines, and deployed applications — across computer vision, healthcare, and legal-tech domains.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript

AI / Deep Learning: PyTorch, Scikit-learn, Transformers, Computer Vision, NLP, Pandas, NumPy

Generative AI & Agentic Systems: LangChain, RAG Pipelines, Vector Search, Sentence Transformers, Groq-hosted LLMs, Prompt Engineering

Backend & Data Systems: FastAPI, MongoDB, ChromaDB, ETL, REST APIs

Cloud & DevOps: Docker, AWS (Fundamentals), Git, CI/CD Concepts

Core CS: Data Structures & Algorithms, Databases, System Design Fundamentals

PROJECTS

MedInsight – Medical Intelligence System | *Python, LangChain, RAG, ChromaDB* GitHub | Live

- Architected a production-grade Retrieval-Augmented Generation platform integrating LangChain, Groq-hosted LLMs, and ChromaDB to power context-aware clinical question answering over biomedical literature.

- Designed the semantic retrieval layer end-to-end — chunking strategy, embedding pipeline, and vector-index tuning — to optimize retrieval precision and response latency across large research corpora.

- Built and deployed a FastAPI backend with a React frontend, applying Biopython for structured extraction and parsing of biological research data.

- Shipped as a fully deployed, publicly accessible platform, demonstrating end-to-end MLOps practice from data ingestion through inference.

DRUNet Image Denoiser | *Python, PyTorch, OpenCV, Transformers* GitHub | Live

- Architected and trained a DRUNet-based deep residual denoising model on DIV2K, applying advanced noise-injection and augmentation strategies to improve generalization across varied noise profiles.

- Engineered an end-to-end PyTorch training pipeline (data loading, augmentation, optimization, checkpointing) enabling reproducible experimentation and rapid iteration.

- Benchmarked performance using PSNR/SSIM against baseline denoising methods and optimized inference for efficient processing of high-resolution images.

- Open-sourced the implementation with a live demo, enabling public evaluation and reuse.

LexTrace – Legal Document Intelligence System | *FastAPI, LangChain, Groq, MongoDB* Live

- Engineered a full-stack RAG pipeline for legal document intelligence, unifying ingestion of scanned and digital documents via PyMuPDF and Tesseract OCR.

- Implemented semantic retrieval using Sentence Transformers and MongoDB vector search, reducing query latency while improving retrieval accuracy across large legal corpora.

- Integrated a Groq-hosted LLM through LangChain to deliver citation-aware legal question answering, prioritizing traceability and factual grounding — a core requirement for enterprise legal-tech systems.

- Built a responsive TypeScript frontend, delivering a production-ready, end-to-end legal-tech application from architecture to UI.

EDUCATION

Vellore Institute of Technology

B.Tech Computer Science & Engineering – CGPA: 8.07

2023–2027

Ideal Public School, Howrah

ISC Class XII Science – 71.08%

2021–2023

ICSE Class X Science – 87%

2013–2021

ACHIEVEMENTS

Meta X PyTorch OpenEnv Hackathon – Finalist

Vellore Institute of Technology

Selected among 800 finalist teams from 31,000+ registered participants nationwide.

2026